

VISHNU S L

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Work Availability: Globally open to any office | Flexible working hours

PROFESSIONAL SUMMARY

Dedicated Transportation Engineer and Traffic Signals & ITS Engineer with a Master of Technology in Transportation Engineering. Proven hands-on experience in traffic safety analysis, congestion management, geospatial data analysis, and laboratory material testing. Experienced in managing intelligent transportation system (ITS) concepts, field data collection campaigns, and geotechnical soil investigations to support infrastructure planning and road safety audits.

CORE SYSTEMS & TECHNICAL SKILLS

- Traffic Simulation & Analysis: PTV VISSIM / PTV Vissim, SUMO, Synchro, Transportation planning / modelling (strategic, operational, & microsimulation approaches)
- GIS & Mapping: ArcGIS Roads and Highways, QGIS, Applications of GIS — highways engineering support
- Design & Drafting: Autodesk AutoCAD Civil 3D, Autodesk AutoCAD, CAD for civil detail, sections, layout & switching drawings
- Data & Analytics Software: Python, Microsoft Power BI, PostgreSQL, MS Office Suite
- Equipment Handling & Surveying: Total Station, DGPS, GPS, Drone Surveying, Vehicle Counting Sensors, Automatic Traffic Counters (ATC), Radar Speed Gun, Loop Detectors, Infrared Sensors, Surveying (topographical, setting out, and control surveying), Survey standards, regulations & topographical survey practice
- Material & Soil Testing: CBR, Universal Testing Machine, Los Angeles Abrasion Test, Geotechnical soil testing (classification, compaction, sieve analysis, Atterberg limits), Direct shear, DCPT
- Bitumen & Pavement Testing: Penetration test, Ductility, Specific gravity, ITS, Marshall stability, Retained Marshall impact, Boiling water test, Viscosity, Water content test
- Core Competencies: Traffic Engineering, Road safety audit (design & operational), Highways operations & maintenance planning, Safety management, Work plan development, Project programming

PROFESSIONAL EXPERIENCE

Arcadis IBI Group Transportation System Engineer / Present

- **Optimized Traffic Signal Timing (Atlanta & Toronto):** Utilized Synchro and SimTraffic to model and optimize complex signal timings for international metropolitan corridors, effectively reducing vehicle delays and queue lengths.
- **Transit Signal Priority (TSP):** Implemented TSP systems across urban corridors to reduce transit delays and improve bus schedule reliability; optimized operations for specific systems like BPRS priority.
- **Digital Twin Development (Bengaluru):** Engineered a comprehensive digital twin model for Bengaluru's transport network to simulate real-time traffic scenarios and enhance large-scale infrastructure planning.
- **Smart Mobility Initiatives:** Supported Advanced Smart Traffic Management (ASTRaM) concepts, including smart mobility planning and Intelligent Transportation System (ITS) integration for Bengaluru.
- **Managed High-Volume Traffic Operations:** Orchestrated parking operations and congestion strategies during critical peak periods, such as the Sabarimala pilgrimage season, ensuring smooth traffic flow and vehicle management.
- **Executed Data-Driven Operational Analysis:** Leveraged pgAdmin and micro-simulation data to identify critical bottlenecks and provided actionable, report-based recommendations to municipal stakeholders.

Accrete Consultant Engineer Assistant Road Safety Engineer / Gurgaon / 2025

- Collected and organised traffic data, site photographs, and field observations received from the audit team.
- Prepared draft and final RSA (Road Safety Audit) reports with clear identification of issues and recommended safety measures.
- Supported formatting, editing, and structuring of reports as per IRC / MoRTH guidelines.

Travancore Geotechnical Investigation and Solution Soil Testing Laboratory Manager / 2025

- Reviewed soil investigation data to support pavement design and infrastructure planning projects.

- Conducted geotechnical soil testing including classification, compaction, CBR, sieve analysis, and Atterberg limits as per relevant standards.

Kerala Highway Research Institute (KHRI) Intern / 2024

- Conducted traffic safety and congestion analysis at Kazhakkootam Junction.
- Collected and analyzed geospatial data using DGPS, Total Station, and Drone Surveying.
- Performed traffic simulations using VISSIM to evaluate congestion patterns.
- Proposed data-driven solutions for road safety improvements and traffic flow optimization.

KSCSTE – NATPAC Intern / Trivandrum / 2022 – 2023

- Conducted soil investigation tests (CBR, direct shear, DCPT) for transportation projects.
- Assisted in pavement testing (Marshall, ITS, retained Marshall impact, boiling water test).
- Collaborated with engineers and researchers in data analysis and interpretation.

College of Engineering Trivandrum Technical Assistant / 2022

- Conducted site investigations, laboratory testing, and field testing.

EDUCATION

- MTech in Transportation Engineering | Rajiv Gandhi Institute of Technology, Kottayam | 2023 – 2025
- BTech in Civil Engineering | PRIST University, Thanjavur | 2018 – 2022
- Diploma in Computer Application | 2016
- Higher Secondary (Plus Two) | 2015
- SSLC | 2013

ACADEMIC PROJECTS

- Optimization of Roundabout Performance in Mixed Traffic Conditions
- Evaluation of Driver Behavior at Signalized Intersections
- Modified Sewage Sludge as Landfill Cover Material
- Variation in Precipitation Characteristics Due to Changes in Land Use/Land Cover Using GIS

TRAINING & WORKSHOPS

- 1st World Symposium on Sustainable Transport and Livability (WSSTL-2025) | Organized by the Department of Civil Engineering, Indian Institute of Science, Bengaluru and the SIG-F4 World Conference on Transport Research Society (WCTRS) on June 24, 2025.
- Internship | Kaiparambu Grama Panchayath, Thrissur.
- International Conference on Recent Scientific Research in Engineering and Technology (ICRSRET2019).

LANGUAGES

- English
- Hindi
- Tamil
- malayalam